Celebel Data Engineering Week 2 Assignment – Using Superstore Dataset(on Kaggle)

Reference: https://www.kaggle.com/datasets/vivek468/superstore-dataset-

Step 1:Load DataSet

1: Create a Database

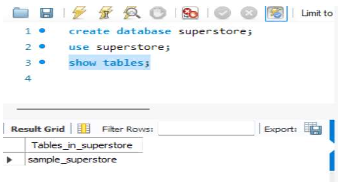
- Created a new database named superstore.

  Command: CREATE DATABASE superstore;

2: Select the Database

- Set the newly created database as the active schema.

  Command: USE superstore;



3: Import the CSV Dataset

- In MySQL Workbench, under the **SCHEMAS** panel, right-click on the superstore database.
- Select **Table Data Import Wizard**.
- Browse and select the SampleSuperstore.csv file downloaded from Kaggle.
- Choose **Create New Table**.
- Enter the table name as sample_superstore.
- Review the detected column names and data types.
- Click **Next** and then **Finish** to import the dataset.

Step 2 : Explore table(schema,sample data)

2.1 — View column structure

Command: DESCRIBE sample_superstore;

```
DESCRIBE sample_superstore;
```

Result Grid | Filter Rows: | Export:

Field	Type	Null	Key	Default	Extra
Ship Mode	text	YES		NULL	
Customer ID	text	YES		NULL	
Customer Name	text	YES		NULL	
Segment	text	YES		NULL	
Country	text	YES		NULL	
City	text	YES		NULL	
State	text	YES		NULL	
Postal Code	int	YES		NULL	
Region	text	YES		NULL	
Product ID	text	YES		NULL	
Category	text	YES		NULL	
Sub-Category	text	YES		NULL	

2.2 — Preview sample rows

Command: SELECT * FROM sample_superstore LIMIT 10;

```
SELECT * FROM sample_superstore LIMIT 10;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code
1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420
2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420
3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036
4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311
5	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311
6	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032
7	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032
8	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032
9	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032
10	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032

2.3 — Total row count

Command: SELECT COUNT(*) AS total_rows FROM sample_superstore;

total_rows
9694

2.4 — Date range of orders

Command: SELECT MIN(`Order_Date`) AS earliest_order, MAX(`Order_Date`) AS latest_order FROM sample_superstore;

earliest_order	latest_order
1/1/2017	9/9/2017

2.5 — Distinct values in key columns

Command: SELECT DISTINCT Region FROM sample_superstore ORDER BY Region;

Result Grid

Region
Central
East
South
West

Command: SELECT DISTINCT Category FROM sample_superstore ORDER BY Category;

Result Grid

Category
Furniture
Office Supplies
Technology

Command: SELECT DISTINCT Segment FROM sample_superstore ORDER BY Segment;

Result Grid

Segment
Consumer
Corporate
Home Office

Command: SELECT DISTINCT `Ship Mode` FROM sample_superstore ORDER BY `Ship Mode`;

Result Grid

Ship Mode
First Class
Same Day
Second Class
Standard Class

Step 3: Apply WHERE filters(region, category, date, sales)

3.1 — Filter by Region

Query: SELECT `Order ID`,` Customer Name`, Category, Sales, Profit

FROM sample_superstore

WHERE Region = 'West'

LIMIT 10;

Result Grid | Filter Rows: | Export: | Wrap Cell Con

Order ID	Customer Name	Category	Sales	Profit
CA-2016-138688	Darrin Van Huff	Office Supplies	14.62	6.8714
CA-2014-115812	Brosina Hoffman	Furniture	48.86	14.1694
CA-2014-115812	Brosina Hoffman	Office Supplies	7.28	1.9656
CA-2014-115812	Brosina Hoffman	Technology	907.152	90.7152
CA-2014-115812	Brosina Hoffman	Office Supplies	18.504	5.7825
CA-2014-115812	Brosina Hoffman	Office Supplies	114.9	34.47
CA-2014-115812	Brosina Hoffman	Furniture	1706.184	85.3092
CA-2014-115812	Brosina Hoffman	Technology	911.424	68.3568
CA-2016-161389	Irene Maddox	Office Supplies	407.976	132.5922
CA-2014-167164	Alejandro Grove	Office Supplies	55.5	9.99

3.2 — Filter by Category

Query: SELECT `Order ID`,` Product Name`, Sales, Profit, Discount

FROM sample_superstore

WHERE Category = 'Technology'

LIMIT 10;

	Order ID	Product Name	Sales	Profit	Discount
▸	CA-2014-115812	Mitel 5320 IP Phone VoIP phone	907.152	90.7152	0.2
	CA-2014-115812	Konftel 250 Conference phone - Charcoal black	911.424	68.3568	0.2
	CA-2014-143336	Cisco SPA 501G IP Phone	213.48	16.011	0.2
	CA-2016-121755	Imation 8GB Mini TravelDrive USB 2.0 Flash Drive	90.57	11.7741	0
	CA-2016-117590	GE 30524EE4	1097.544	123.4737	0.2
	CA-2015-117415	Plantronics HL10 Handset Lifter	371.168	41.7564	0.2
	CA-2017-120999	Panasonic Kx-TS550	147.168	16.5564	0.2
	CA-2016-118255	Verbatim 25 GB 6x Blu-ray Single Layer Recorda...	45.98	19.7714	0
	CA-2016-169194	Imation 8gb Micro Traveldrive Usb 2.0 Flash Drive	45	4.95	0
	CA-2016-169194	LF Elite 3D Dazzle Designer Hard Case Cover, Lf...	21.8	6.104	0

3.3 — Filter by date range

Query: SELECT \`Order ID\`,\` Order Date\`, \`Customer Name\`, Sales

FROM sample_superstore

WHERE \`Order Date\` BETWEEN '2017-01-01' AND '2017-12-31'

ORDER BY \`Order Date\`

LIMIT 10;

	Order ID	Order Date	Customer Name	Sales

3.4 — Filter high-value orders (Sales > 500)

Query: SELECT \`Order ID\`,\` Customer Name\`, \`Product Name\`, Sales, Profit

FROM sample_superstore

WHERE Sales > 500

ORDER BY Sales DESC

LIMIT 10;

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

	Order ID	Customer Name	Product Name	Sales	Profit
▸	CA-2014-145317	Sean Miller	Cisco TelePresence System EX90 Videoconferen...	22638.48	-1811.
	CA-2016-118689	Tamara Chand	Canon imageCLASS 2200 Advanced Copier	17499.95	8399.9
	CA-2017-140151	Raymond Buch	Canon imageCLASS 2200 Advanced Copier	13999.96	6719.9
	CA-2017-127180	Tom Ashbrook	Canon imageCLASS 2200 Advanced Copier	11199.968	3919.9
	CA-2017-166709	Hunter Lopez	Canon imageCLASS 2200 Advanced Copier	10499.97	5039.9
	CA-2016-117121	Adrian Barton	GBC Ibimaster 500 Manual ProClick Binding System	9892.74	4946.3
	CA-2014-116904	Sanjit Chand	Ibico EPK-21 Electric Binding System	9449.95	4630.4
	US-2016-107440	Bill Shonely	3D Systems Cube Printer, 2nd Generation, Mag...	9099.93	2365.9
	CA-2016-158841	Sanjit Engle	HP Designjet T520 Inkjet Large Format Printer -...	8749.95	2799.9
	CA-2016-143714	Christopher Conant	Canon imageCLASS 2200 Advanced Copier	8399.976	1119.9

Step 4: Use GROUP BY for aggregations(sales, quantity, averages)

4.1 — Sales by Category

Query: SELECT

Category,

COUNT(*) AS order_count,

ROUND(SUM(Sales), 2) AS total_sales,

ROUND(AVG(Sales), 2) AS avg_order_value,

ROUND(SUM(Profit), 2) AS total_profit

FROM sample_superstore

GROUP BY Category

ORDER BY total_sales DESC;

Category	order_count	total_sales	avg_order_value	total_profit
Technology	1839	835900.07	454.54	145387.1
Furniture	2074	733046.86	353.45	16980.77
Office Supplies	5781	703502.93	121.69	120489.89

4.2 — Sales by Region

Query: SELECT

Region,

COUNT(*) AS order_count,

ROUND(SUM(Sales), 2) AS total_sales,

ROUND(SUM(Profit), 2) AS total_profit,

ROUND(AVG(Discount)*100, 1) AS avg_discount_pct

FROM sample_superstore

GROUP BY Region

ORDER BY `total_sales` DESC;

Region	order_count	total_sales	total_profit	avg_discount_pct
West	3099	713471.34	106021.15	11
East	2756	672194.05	90672.01	14.3
Central	2260	497800.87	40128.9	23.8
South	1579	388983.59	46035.69	14.5

4.3 — Quantity and sales by Sub-Category

Query: SELECT

Sub_Category,

SUM(Quantity) AS total_quantity,

ROUND(SUM(Sales), 2) AS total_sales,

ROUND(AVG(Sales), 2)   AS avg_order_value,

ROUND(SUM(Profit), 2) AS total_profit

FROM sample_superstore

GROUP BY Sub_Category

ORDER BY total_sales DESC;

Region	order_count	total_sales	total_profit	avg_discount_pct
West	3099	713471.34	106021.15	11
East	2756	672194.05	90672.01	14.3
Central	2260	497800.87	40128.9	23.8
South	1579	388983.59	46035.69	14.5

4.4 — Sales by Segment

Query: SELECT

Segment,

COUNT(*) AS order_count,

ROUND(SUM(Sales), 2)   AS total_sales,

ROUND(SUM(Profit), 2) AS total_profit,

ROUND(AVG(Sales), 2)   AS avg_order_value

FROM sample_superstore

GROUP BY Segment

ORDER BY total_sales DESC;

Segment	order_count	total_sales	total_profit	avg_order_value
Consumer	5044	1150166.18	132669.78	228.03
Corporate	2920	696604.51	90366.3	238.56
Home Office	1730	425679.16	59821.68	246.06

4.5 — Region + Category combined grouping

Query: SELECT

Region,

Category,

ROUND(SUM(Sales), 2)   AS total_sales,

ROUND(SUM(Profit), 2) AS total_profit

FROM sample_superstore

GROUP BY Region, Category

ORDER BY Region, total_sales DESC;

	Region	Category	total_sales	total_profit
▸	Central	Technology	170401.53	33693.44
	Central	Office Supplies	164616.2	9020.72
	Central	Furniture	162783.14	-2585.25
	East	Technology	264872.08	47439.56
	East	Furniture	205540.35	2501.82
	East	Office Supplies	201781.62	40730.64
	South	Technology	148730.52	19982.82
	South	Office Supplies	123979.92	19577.08
	South	Furniture	116273.14	6475.79
	West	Technology	251895.93	44271.28
	West	Furniture	248450.23	10588.42
	West	Office Supplies	213125.18	51161.45

Step 5: Sort and limit results(top products, top categories)

5.1 — Top 5 Sub-Categories by revenue

Query: SELECT

`Sub-Category`,

ROUND(SUM(Sales), 2) AS revenue

FROM sample_superstore

GROUP BY `Sub-Category`

ORDER BY revenue DESC

LIMIT 5;

	Region	Category	total_sales	total_profit
▸	Central	Technology	170401.53	33693.44
	Central	Office Supplies	164616.2	9020.72
	Central	Furniture	162783.14	-2585.25
	East	Technology	264872.08	47439.56
	East	Furniture	205540.35	2501.82
	East	Office Supplies	201781.62	40730.64
	South	Technology	148730.52	19982.82
	South	Office Supplies	123979.92	19577.08
	South	Furniture	116273.14	6475.79
	West	Technology	251895.93	44271.28
	West	Furniture	248450.23	10588.42
	West	Office Supplies	213125.18	51161.45

5.2 — Bottom 5 Sub-Categories by revenue

Query: SELECT

`Sub-Category`,

ROUND(SUM(Sales), 2) AS revenue

FROM sample_superstore

GROUP BY `Sub-Category`

ORDER BY revenue ASC

LIMIT 5;

	Sub-Category	revenue
▸	Fasteners	3008.66
	Labels	12486.31
	Envelopes	15339.49
	Art	27118.79
	Supplies	45952.47

5.3 — Top 10 products by total sales

```
Query: SELECT
    `Product Name`,
    ROUND(SUM(Sales), 2)  AS total_sales,
    ROUND(SUM(Profit), 2) AS total_profit
    FROM sample_superstore
    GROUP BY `Product Name`
    LIMIT 10;
```

Product Name	total_sales	total_profit
Bush Somerset Collection Bookcase	1263.96	-56.32
Hon Deluxe Fabric Upholstered Stacking Chairs,...	10637.53	1927.44
Self-Adhesive Address Labels for Typewriters b...	153.51	64.4
Bretford CR4500 Series Slim Rectangular Table	7242.77	-532.76
Eldon Fold 'N Roll Cart System	301.97	63.75
Eldon Expressions Wood and Plastic Desk Acces...	160.54	46.56
Newell 322	30.58	6.66
Mitel 5320 IP Phone VoIP phone	2116.69	211.67
DXL Angle-View Binders with Locking Rings by S...	215.11	66.69
Belkin F5C206VTEL 6 Outlet Surge	464.2	110.3

5.4 — Top 5 most profitable products

```
Query: SELECT
    `Product Name`,
    ROUND(SUM(Profit), 2) AS total_profit,
    ROUND(SUM(Sales), 2)  AS total_sales
    FROM sample_superstore
    GROUP BY `Product Name`
    ORDER BY total_profit DESC
    LIMIT 5;
```

Product Name	total_profit	total_sales
Canon imageCLASS 2200 Advanced Copier	25199.93	61599.82
Fellowes PB500 Electric Punch Plastic Comb Bind...	7753.04	27453.38
Hewlett Packard LaserJet 3310 Copier	6983.88	18839.69
Canon PC1060 Personal Laser Copier	4570.93	11619.83
HP Designjet T520 Inkjet Large Format Printer -...	4094.98	18374.9

5.5 — Top 5 states by total sales

```
Query: SELECT
    State,
    ROUND(SUM(Sales), 2)  AS total_sales,
    ROUND(SUM(Profit), 2) AS total_profit
```

FROM sample_superstore

GROUP BY State

ORDER BY total_sales DESC

LIMIT 5;

	State	total_sales	total_profit
▶	California	450567.59	74669.2
	New York	309453.63	73507.13
	Texas	169553.63	-25534.99
	Washington	136590.17	32976.62
	Pennsylvania	114911.24	-15446.38

Step 6: Solve use cases (monthly trends, top cutomers, duplicates)

6.1 — Monthly sales trend

Query: SELECT

DATE_FORMAT(`Order Date`, '%Y-%m') AS month,

COUNT(*) AS order_count,

ROUND(SUM(Sales), 2) AS monthly_sales,

ROUND(SUM(Profit), 2) AS monthly_profit

FROM sample_superstore

GROUP BY month

ORDER BY month;

	month	order_count	monthly_sales	monthly_profit
▶	NULL	9694	2272449.86	282857.75

6.2 — Yearly sales trend

Query: SELECT

YEAR(Order_Date)   AS year,

COUNT(*) AS order_count,

ROUND(SUM(Sales), 2) AS yearly_sales,

ROUND(SUM(Profit), 2) AS yearly_profit

FROM sample_superstore

GROUP BY year

ORDER BY year;

year	order_count	yearly_sales	yearly_profit
NULL	9694	2272449.86	282857.75

6.3 — Top 10 customers by revenue

Query: SELECT

Customer_Name,

Segment,

COUNT(*) AS total_orders,

ROUND(SUM(Sales), 2) AS total_spent,

ROUND(SUM(Profit), 2) AS profit_generated

FROM sample_superstore

GROUP BY Customer_Name, Segment

ORDER BY total_spent DESC

LIMIT 10;

Customer Name	Segment	total_orders	total_spent	profit_generated
Sean Miller	Home Office	15	25043.05	-1980.74
Tamara Chand	Corporate	11	19017.85	8964.48
Raymond Buch	Consumer	18	15117.34	6976.1
Tom Ashbrook	Home Office	10	14595.62	4703.79
Adrian Barton	Consumer	19	14355.61	5438.91
Sanjit Chand	Consumer	22	14142.33	5757.41
Ken Lonsdale	Consumer	27	14071.92	768.87
Hunter Lopez	Consumer	11	12873.3	5622.43
Sanjit Engle	Consumer	19	12209.44	2650.68
Christopher Conant	Consumer	11	12129.07	2177.05

6.4 — Shipping speed by Ship Mode

Query :SELECT

`Ship Mode`,

COUNT(*) AS shipment_count,

ROUND(AVG(DATEDIFF(`Ship Date`, `Order Date`)), 1)

AS avg_days_to_ship

FROM sample_superstore

GROUP BY `Ship Mode`

ORDER BY avg_days_to_ship;

	Ship Mode	shipment_count	avg_days_to_ship
▶	Second Class	1886	NULL
	Standard Class	5780	NULL
	First Class	1501	NULL
	Same Day	527	NULL

6.5 — Profit margin by Category

Query: SELECT

Category,

ROUND(SUM(Sales), 2) AS total_sales,

ROUND(SUM(Profit), 2) AS total_profit,

ROUND((SUM(Profit)/SUM(Sales))*100, 2)   AS margin_pct

FROM sample_superstore

GROUP BY Category

ORDER BY margin_pct DESC;

	Category	total_sales	total_profit	margin_pct
▶	Technology	835900.07	145387.1	17.39
	Office Supplies	703502.93	120489.89	17.13
	Furniture	733046.86	16980.77	2.32

Step 7: Validate results (row counts, data quality).

7.1 — Confirm total row count

Query: SELECT COUNT(*) AS total_rows

FROM sample_superstore;

	total_rows
▶	9694

7.2 — Check for NULL values in key columns

Query : SELECT

SUM(CASE WHEN `Order ID` IS NULL THEN 1 ELSE 0 END) AS null_order_id,

SUM(CASE WHEN `Customer Name` IS NULL THEN 1 ELSE 0 END) AS null_customer,

SUM(CASE WHEN Sales IS NULL THEN 1 ELSE 0 END) AS null_sales,

SUM(CASE WHEN Profit IS NULL THEN 1 ELSE 0 END) AS null_profit,

SUM(CASE WHEN `Order Date`   IS NULL THEN 1 ELSE 0 END) AS null_date,

SUM(CASE WHEN Region   IS NULL THEN 1 ELSE 0 END) AS null_region

FROM sample_superstore;

7.3 — Check for invalid Sales or Discount values

	null_order_id	null_customer	null_sales	null_profit	null_date	null_region
▶	0	0	0	0	0	0

7.4 — Verify regional totals match grand total

Query: SELECT

(SELECT ROUND(SUM(Sales),2) FROM sample_superstore)

AS grand_total,

SUM(regional_sales) AS sum_of_regions

FROM (

SELECT Region, ROUND(SUM(Sales),2) AS regional_sales

FROM sample_superstore

GROUP BY Region

) AS region_breakdown;

	grand_total	sum_of_regions
▶	2272449.86	2272449.8499999996

7.5 — Check for impossible ship dates

Query: SELECT COUNT(*) AS bad_dates

FROM sample_superstore

WHERE `Ship Date` < `Order Date`;

	bad_dates
▶	1519

7.6 — Summary of unique counts

Query: SELECT

COUNT(DISTINCT `Order ID`) AS unique_orders,

COUNT(DISTINCT `Customer ID`) AS unique_customers,

COUNT(DISTINCT `Product ID`) AS unique_products,

COUNT(DISTINCT City) AS unique_cities,

COUNT(DISTINCT State) AS unique_states

```
FROM sample_superstore;
```

	unique_orders	unique_customers	unique_products	unique_cities	unique_states
▶	4931	793	1812	529	49